

The Union Formula for (I) -Envelopes: Verification, Literature Status, and an Affine-Union Theorem

Research note prepared from Question 5.1 of Kalenda–Raja

June 22, 2026

Abstract

Kalenda and Raja asked whether

$$(\text{I})\text{env}(A \cup B) = (\text{I})\text{env}(A) \cup (\text{I})\text{env}(B)$$

whenever $A, B \subset X^*$ are convex and $A \cup B$ is convex. We first check the proposed partial argument. It is correct, but its topological part is already exactly the implication $(1) \Rightarrow (2)$ in their Theorem 4.3; combined with their Theorem 3.3 it gives the formula when X contains no isomorphic copy of ℓ^1 . A literature search through June 22, 2026 did not locate a full proof or a counterexample, and the current arXiv abstract still describes the underlying non-locally-convex topological problem as open.

We then record further positive results. The formula is automatic when the union is already (I) -closed (in particular, when it is $\sigma(X^*, B_1(X))$ -closed), and it holds whenever $A \cup B$ is norm separable. More substantially, it holds in every Banach space whenever $A \cup B$ is an affine subspace of X^* (not necessarily closed). The latter is proved from the sequential separation criterion for (I) -envelopes and a self-contained geometric lemma: two proper closed convex sets covering a real Banach space must be two oppositely oriented parallel half-spaces, possibly with an overlapping slab. Consequently, any counterexample to Question 5.1 must occur in a space containing ℓ^1 , must have a non-norm-separable union, and cannot have affine union.

1 The question and the outcome of the check

Let X be a real or complex Banach space. For $S \subset X^*$, its (I) -envelope is

$$(\text{I})\text{env}(S) = \bigcap \left\{ \overline{\text{co} \left(\bigcup_{n=1}^{\infty} \overline{\text{co } S_n}^{w^*} \right)}^{\|\cdot\|} : S = \bigcup_{n=1}^{\infty} S_n \right\}.$$

The open problem is the following [1, Question 5.1].

Union question. If $A, B \subset X^*$ are convex and $A \cup B$ is convex, must

$$(\text{I})\text{env}(A \cup B) = (\text{I})\text{env}(A) \cup (\text{I})\text{env}(B) \tag{UF}$$

hold?

The inclusion from right to left always holds, by monotonicity of the (I) -envelope. The issue is the reverse inclusion.

5. FINAL REMARKS AND OPEN PROBLEMS

The answer to the ‘locally convex’ variant of Question 1.1 is for some Banach spaces positive and for some Banach spaces negative as witnessed by the results of Section 3. However, a characterization of those Banach spaces for which the answer is positive is still missing as pointed in Question 3.9.

The ‘topological’ variant of Question 1.1 is widely open. We have no counterexample to it, but we do not have either any example for which the ‘topological’ variant has a positive answer but the ‘locally convex’ variant has a negative answer. In view of Theorem 4.3 the following problem is natural to ask:

Question 5.1. *Assume that X is a Banach space and $A, B \subset X^*$ are two convex sets such that $A \cup B$ is also convex. Is $(\text{I-env}(A \cup B) = (\text{I-env}(A) \cup (\text{I-env}(B))$?*

Further, Question 3.10 suggests that there may be a substantial difference between (I)-envelopes of bounded and unbounded sets. It seems that the following question is interesting and open.

This packet is a **strengthened partial packet**. It supersedes the earlier June 14 packet, which recorded only the topological/no- ℓ^1 subcase. The earlier version is preserved in a dated **history**/subfolder, and the submitted June 22 report is preserved in a dated **evidence**/subfolder. The general case of Question 5.1 is still not claimed to be solved.

Proposition 1.1 (The proposed topological argument is correct). *Suppose there is a topology τ on X^* such that*

$$(\mathbf{I})\text{env}(S) = \overline{\text{co}} S^\tau \quad (S \subset X^*).$$

Then (UF) holds for every admissible pair A, B . In particular, (UF) holds if X contains no isomorphic copy of ℓ^1 .

Proof. Put $C = A \cup B$. Since A, B, C are convex,

$$(\mathbf{I})\text{env}(C) = \overline{C}^\tau = \overline{A \cup B}^\tau = \overline{A}^\tau \cup \overline{B}^\tau = (\mathbf{I})\text{env}(A) \cup (\mathbf{I})\text{env}(B).$$

If X contains no copy of ℓ^1 , Kalenda–Raja’s Theorem 3.3 gives, for every $S \subset X^*$,

$$(\mathbf{I})\text{env}(S) = \overline{\text{co}} S^{\sigma(X^*, B_1(X))},$$

so the first part applies. □

Remark 1.2 (Novelty of the submitted partial result). The first assertion of Proposition 1.1 is not a new lemma: it is precisely the proof of $(1) \Rightarrow (2)$ in [1, Theorem 4.3]. The “no ℓ^1 ” conclusion is the immediate combination of that implication with [1, Theorem 3.3]. The submitted partial result is therefore mathematically correct, including its use of Theorem 3.3 for arbitrary subsets, but it should be presented as a useful corollary already contained in the source paper rather than as a novel result.

2 Literature status as of June 22, 2026

The arXiv record now lists the journal publication *Topology and its Applications* **364** (2025), Article 109103, 27 pages. The arXiv abstract still says that the version without local convexity is open. Searches by the exact paper title, DOI/journal data, Question 5.1, the displayed union formula, “(I)-envelope”, and the notation cl_{IF} found the source paper and bibliographic records, but no later paper announcing either a proof or a counterexample.

This is not a proof that no unpublished solution exists. It does support the following conservative status statement:

No full answer or counterexample was located in the indexed literature through June 22, 2026; Question 5.1 should still be treated as open unless a newer source is supplied.

The remainder of this note develops additional rigorous positive cases.

3 Sequential separation and elementary invariances

All convex and affine notions below are over $\mathbb{R}$. In the complex case we use real parts, exactly as in [1, Lemma 2.1].

For a bounded sequence $\mathbf{x} = (x_n) \subset B_X$, set

$$p_{\mathbf{x}}(x^*) = \limsup_{n \rightarrow \infty} \text{Re } x^*(x_n), \quad x^* \in X^*.$$

This is a finite, positively homogeneous, subadditive, norm-Lipschitz function.

Lemma 3.1 (Sequential separation criterion [2, 1]). *Let $S \subset X^*$ and $\eta \in X^*$. Then $\eta \notin (\text{I})\text{env}(S)$ if and only if there is a sequence $(x_n) \subset B_X$ such that*

$$\sup_{s \in S} \limsup_{n \rightarrow \infty} \text{Re } s(x_n) < \inf_{n \in \mathbb{N}} \text{Re } \eta(x_n). \quad (3.1)$$

Equivalently, one may replace the right-hand side by $\liminf_n \text{Re } \eta(x_n)$ or by $\limsup_n \text{Re } \eta(x_n)$.

We will repeatedly use two simple consequences.

Lemma 3.2 (Norm closure and translation). *For every $S \subset X^*$ and $u \in X^*$,*

$$(\text{I})\text{env}(S) = (\text{I})\text{env}(\overline{S}^{\|\cdot\|}), \quad (\text{I})\text{env}(S + u) = (\text{I})\text{env}(S) + u.$$

Proof. For the first assertion, p_x is norm continuous and in fact

$$|p_x(x^*) - p_x(y^*)| \leq \|x^* - y^*\|.$$

Thus its supremum on S equals its supremum on the norm closure of S , and Lemma 3.1 gives the equality of the envelopes. Translation invariance follows directly from the defining countable-cover formula: translation commutes with convex hulls, weak-star closure, norm closure, and intersections. $\square$

4 Further immediate positive cases

Proposition 4.1 ((I)-closed union). *Let $A, B \subset X^*$ be convex and let $C = A \cup B$ be convex. If $(\text{I})\text{env}(C) = C$, then (UF) holds. In particular, this is true whenever C is $\sigma(X^*, B_1(X))$ -closed, and hence whenever C is weak-star closed.*

Proof. Every point of $C = A \cup B$ belongs to $(\text{I})\text{env}(A) \cup (\text{I})\text{env}(B)$. Conversely, monotonicity gives

$$(\text{I})\text{env}(A) \cup (\text{I})\text{env}(B) \subset (\text{I})\text{env}(C) = C.$$

For the sufficient condition, Kalenda–Raja’s Proposition 2.2 gives

$$(\text{I})\text{env}(S) \subset \overline{\text{co}} S^{\sigma(X^*, B_1(X))} \quad (S \subset X^*).$$

Thus a convex $\sigma(X^*, B_1(X))$ -closed set C satisfies $(\text{I})\text{env}(C) = C$. Since $X \subset B_1(X)$, weak-star closedness is an even stronger sufficient condition for a convex set. $\square$

Proposition 4.2 (Norm-separable union). *If $A, B \subset X^*$ are convex, $A \cup B$ is convex, and $A \cup B$ is norm separable, then (UF) holds.*

Proof. For every norm-separable set $S \subset X^*$ one has

$$(\text{I})\text{env}(S) = \overline{\text{co}} S^{\|\cdot\|};$$

see [2, Remark 1.1(ii)] or [1, Proposition 4.4(b)]. Since A , B , and $C = A \cup B$ are convex and norm separable,

$$(\text{I})\text{env}(C) = \overline{C}^{\|\cdot\|} = \overline{A \cup B}^{\|\cdot\|} = \overline{A}^{\|\cdot\|} \cup \overline{B}^{\|\cdot\|} = (\text{I})\text{env}(A) \cup (\text{I})\text{env}(B).$$

$\square$

5 Geometry of a two-set convex cover

The key geometric input for the affine-union theorem is elementary but useful in its own right.

For a nonempty closed convex set P in a real Banach space V , write

$$\text{rec}(P) = \{v \in V : P + [0, \infty)v \subset P\}$$

for its recession cone.

Lemma 5.1 (One ray determines a recession direction). *Let $P \subset V$ be nonempty, closed, and convex. If $p_0 + tv \in P$ for all $t \geq 0$ for one $p_0 \in P$, then $v \in \text{rec}(P)$.*

Proof. Fix $p \in P$ and $t \geq 0$. For $s > t$, convexity gives

$$\left(1 - \frac{t}{s}\right)p + \frac{t}{s}(p_0 + sv) = p + tv + \frac{t}{s}(p_0 - p) \in P.$$

Letting $s \rightarrow \infty$ and using closedness yields $p + tv \in P$. $\square$

Lemma 5.2 (Two closed convex cones covering a Banach space). *Let C, D be proper closed convex cones in a real Banach space V , and suppose $C \cup D = V$. Then there is a nonzero $g \in V^*$ such that, after interchanging C and D if necessary,*

$$C = \{v : g(v) \leq 0\}, \quad D = \{v : g(v) \geq 0\}.$$

Proof. We first claim that

$$x \in C \setminus D \implies -x \in D \setminus C. \quad (5.1)$$

Indeed, if also $-x \in C$, then the whole line $\mathbb{R}x$ lies in C . Choose $y \in D \setminus C$, which is possible because C is proper and $C \cup D = V$. For every $t \in \mathbb{R}$, the point $y + tx$ cannot lie in C ; otherwise $y = (y + tx) - tx \in C$. Hence $y + tx \in D$ for every t . For $t > 0$, $(y + tx)/t \in D$, and closedness gives $x \in D$ as $t \rightarrow \infty$, a contradiction. This proves (5.1); the symmetric statement holds with C, D interchanged.

Set $H = C \cap D$. It is a closed linear subspace. It is already a convex cone; if $h \in H$ and $-h$ belonged to exactly one of C, D , then (5.1) would force h to belong to exactly the other one, a contradiction. Thus $-h \in H$.

Let $q : V \rightarrow V/H$ be the quotient map. Since $C + H = C$ and $D + H = D$, the sets $q(C)$ and $q(D)$ are closed convex cones in V/H . They cover V/H and intersect only at 0. Moreover, by (5.1), $q(D) = -q(C)$. Consequently

$$(V/H) \setminus \{0\} = (q(C) \setminus \{0\}) \dot{\cup} (q(D) \setminus \{0\})$$

is a separation into two nonempty relatively closed sets. A real normed space of dimension at least two remains path connected after deleting the origin: join two nonzero points by a segment unless they are opposite, and in the opposite case use an intermediate vector outside their line. Hence $\dim(V/H) = 1$.

Thus H is a closed hyperplane. Choose a nonzero $g \in V^*$ with $H = \ker g$. The two quotient cones are the two half-lines in $\mathbb{R}$, so after reversing the sign of g if needed, the asserted formulas follow. $\square$

Proposition 5.3 (Slab structure). *Let P, Q be proper nonempty closed convex subsets of a real Banach space V , and suppose $P \cup Q = V$. Then there are a nonzero $g \in V^*$ and numbers $\beta \leq \alpha$ such that, after interchanging P, Q if necessary,*

$$P = \{v \in V : g(v) \leq \alpha\}, \quad Q = \{v \in V : g(v) \geq \beta\}. \quad (5.2)$$

Proof. Because V is connected and P, Q are closed proper sets covering V , one has $P \cap Q \neq \emptyset$. Translate by a point of the intersection, so that $0 \in P \cap Q$.

Fix $v \in V$. The sets

$$I_P = \{t \geq 0 : tv \in P\}, \quad I_Q = \{t \geq 0 : tv \in Q\}$$

are closed intervals containing 0, and $I_P \cup I_Q = [0, \infty)$. At least one is unbounded, hence equals $[0, \infty)$. By Lemma 5.1, $v \in \text{rec}(P) \cup \text{rec}(Q)$. Therefore

$$\text{rec}(P) \cup \text{rec}(Q) = V.$$

Neither recession cone is all of V , since $0 \in P \cap Q$ and both P, Q are proper. Lemma 5.2 supplies a nonzero $g \in V^*$ such that

$$\text{rec}(P) = \{g \leq 0\}, \quad \text{rec}(Q) = \{g \geq 0\}.$$

In particular, $H = \ker g$ is contained in the lineality space of each set, so $P + H = P$ and $Q + H = Q$.

Hence membership in P depends only on the scalar $g(v)$. The set $g(P)$ is a nonempty closed convex subset of $\mathbb{R}$ (closedness follows because P is saturated with respect to the quotient map $V \rightarrow V/H$). It is downward closed, because every direction with negative g -value belongs to $\text{rec}(P)$. Since P is proper, $g(P) = (-\infty, \alpha]$ for some finite α . Similarly, $g(Q) = [\beta, \infty)$ for some finite β . The covering property gives $\beta \leq \alpha$, proving (5.2). $\square$

6 The affine-union theorem

Theorem 6.1 (Affine-union formula). *Let X be any real or complex Banach space. Let $A, B \subset X^*$ be convex, and suppose that $A \cup B$ is an affine subspace of the underlying real vector space of X^* . Then*

$$(\text{I})\text{env}(A \cup B) = (\text{I})\text{env}(A) \cup (\text{I})\text{env}(B).$$

The affine subspace need not be norm closed.

Proof. By Lemma 3.2, replacing A, B and $A \cup B$ by their norm closures does not change any of the three (I) -envelopes. Since norm closure commutes with finite unions,

$$\overline{A \cup B}^{\|\cdot\|} = \overline{A}^{\|\cdot\|} \cup \overline{B}^{\|\cdot\|},$$

and the closure of an affine subspace is affine. Translation invariance therefore reduces the problem to the following situation: $V \subset X^*$ is a norm-closed real linear subspace, and $P, Q \subset V$ are closed convex sets with $P \cup Q = V$.

If $P = V$ or $Q = V$, the result is immediate. Otherwise Proposition 5.3 gives $g \in V^* \setminus \{0\}$ and $\beta \leq \alpha$ such that

$$P = \{v \in V : g(v) \leq \alpha\}, \quad Q = \{v \in V : g(v) \geq \beta\}.$$

Choose $u \in V$ with $g(u) = 1$, and put $H = \ker g$.

We prove the nontrivial inclusion by contraposition. Suppose

$$\eta \notin (\text{I})\text{env}(P) \cup (\text{I})\text{env}(Q).$$

Apply Lemma 3.1 to P . There is $(x_n) \subset B_X$ with

$$M_P := \sup_{p \in P} \limsup_n \text{Re } p(x_n) < \inf_n \text{Re } \eta(x_n).$$

Write $p_{\mathbf{x}}(v) = \limsup_n \operatorname{Re} v(x_n)$. Fix $p_0 \in P$. For every $h \in H$ and $t > 0$, $p_0 \pm th \in P$, hence subadditivity gives

$$tp_{\mathbf{x}}(\pm h) = p_{\mathbf{x}}(\pm th) \leq p_{\mathbf{x}}(p_0 \pm th) + p_{\mathbf{x}}(-p_0) \leq M_P + p_{\mathbf{x}}(-p_0).$$

Letting $t \rightarrow \infty$ shows $p_{\mathbf{x}}(h) \leq 0$ and $p_{\mathbf{x}}(-h) \leq 0$; since $0 \leq p_{\mathbf{x}}(h) + p_{\mathbf{x}}(-h)$, both are zero. Thus

$$\operatorname{Re} h(x_n) \longrightarrow 0 \quad (h \in H). \quad (6.1)$$

Also $p_0 - tu \in P$ for $t \geq 0$, so the same argument gives $p_{\mathbf{x}}(-u) \leq 0$, or $\liminf_n \operatorname{Re} u(x_n) \geq 0$.

Pass to a subsequence on which

$$\operatorname{Re} u(x_n) \rightarrow r \geq 0, \quad \operatorname{Re} \eta(x_n) \rightarrow s.$$

For $p \in P$, the vector $p - g(p)u$ lies in H , so by (6.1)

$$\operatorname{Re} p(x_n) \longrightarrow rg(p).$$

Consequently the supremum of the new limsups over P is $r\alpha$. Passing to a subsequence cannot increase any individual limsup, and its η -limit is at least the old infimum; hence

$$s > r\alpha. \quad (6.2)$$

Apply the same argument to a separating sequence $(y_n) \subset B_X$ for Q . Since H is again the lineality space while $u \in \operatorname{rec}(Q)$, after passing to a subsequence we obtain

$$\operatorname{Re} h(y_n) \rightarrow 0 \quad (h \in H), \quad \operatorname{Re} u(y_n) \rightarrow t \leq 0, \quad \operatorname{Re} \eta(y_n) \rightarrow v > t\beta. \quad (6.3)$$

If $r = 0$, then (6.1) and the decomposition $c = g(c)u + (c - g(c)u)$ show that $\operatorname{Re} c(x_n) \rightarrow 0$ for every $c \in V$, whereas $s > 0$ by (6.2). Lemma 3.1 gives $\eta \notin (\operatorname{I})\operatorname{env}(V)$. The case $t = 0$ follows similarly from (6.3).

It remains to consider $r > 0$ and $t < 0$. Define

$$z_n = (-t)x_n + ry_n, \quad w_n = \frac{z_n}{-t + r} \in B_X.$$

For each $c \in V$, using $c = g(c)u + (c - g(c)u)$ and (6.1)–(6.3),

$$\operatorname{Re} c(z_n) \longrightarrow (-t)rg(c) + rtg(c) = 0.$$

On the other hand, by (6.2) and (6.3),

$$\lim_n \operatorname{Re} \eta(z_n) = (-t)s + rv > (-t)r\alpha + rt\beta = rt(\beta - \alpha) \geq 0,$$

because $r > 0$, $t < 0$, and $\beta \leq \alpha$. After division by $-t + r$, the η -limit remains strictly positive, while every element of V has real part tending to zero on (w_n) . Lemma 3.1 again yields $\eta \notin (\operatorname{I})\operatorname{env}(V)$.

We have proved

$$\eta \notin (\operatorname{I})\operatorname{env}(P) \cup (\operatorname{I})\operatorname{env}(Q) \implies \eta \notin (\operatorname{I})\operatorname{env}(V),$$

which is the desired reverse inclusion. The forward inclusion follows from monotonicity. $\square$

Corollary 6.2. *No counterexample to Question 5.1 can have $A \cup B$ equal to a linear subspace, a hyperplane, or more generally any affine subspace of X^* . In particular, the tempting strategy of splitting a hyperplane $\ker x^{**}$ into two convex pieces cannot produce a counterexample, even when $x^{**} \notin B_1(X)$ and $(\operatorname{I})\operatorname{env}(\ker x^{**}) = X^*$.*

7 What a counterexample would now have to look like

Combining the source theorem with Propositions 4.1 and 4.2 and Theorem 6.1 gives four necessary conditions for a counterexample:

- (i) X must contain an isomorphic copy of ℓ^1 ;
- (ii) $A \cup B$ must be non-norm-separable;
- (iii) $A \cup B$ must be genuinely non-affine;
- (iv) $A \cup B$ must fail to be (I) -closed (in particular, it cannot be $\sigma(X^*, B_1(X))$ -closed or weak-star closed).

These restrictions eliminate several of the most natural constructions based on separable subspaces, hyperplanes, or opposing half-spaces.

It is useful to spell out the remaining technical obstruction. Given a bounded sequence $(x_n) \subset X$, let K be its weak-star cluster set in $B_{X^{**}}$. Then

$$p_X(x^*) = \max_{x^{**} \in K} \operatorname{Re} x^{**}(x^*).$$

A strong separating sequence for A therefore produces a compact cluster set all of whose points separate η from A with a uniform gap; a second sequence does the analogous job for B . To separate from $A \cup B$, one would like a sequence whose cluster set lies in the intersection of the two separator regions. In a nonmetrizable weak-star compact space, however, cluster sets generated by sequences are not stable under arbitrary intersections. This is the point at which the obvious “combine the two witnesses” argument breaks down. The affine proof above works because the slab geometry forces the restrictions of the two sequences to be one-dimensional, allowing an explicit cancellation.

The general convex-union problem has no comparable one-dimensional reduction in the argument currently known here. Thus the results above are strictly partial and do not settle Question 5.1.

8 Conclusion

The submitted proof is valid but already contained in the source paper’s Theorem 4.3, with its no- ℓ^1 consequence supplied by Theorem 3.3. No full solution or counterexample was located in the literature search. The newest rigorous progress obtained here is the affine-union theorem, together with the norm-separable and (I) -closed union observations. These results narrow any future counterexample search to a nonseparable, non-affine convex union in the dual of a Banach space containing ℓ^1 .

References

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- [2] O. F. K. Kalenda, *(I) -envelopes of closed convex sets in Banach spaces*, Israel J. Math. **162** (2007), 157–181.